



37000: Marketing Strategy

Advertising and Evaluation

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Why might TV ad effects be particularly small?

- Is it a “law of nature” that ad ROI is small?

Alternatively TV is:

- Far from the point of purchase.
- Difficult to target precisely.
- Expensive

These would suggest that we should be able to do better online, right?

THE VALUE OF OFFSITE DATA TO ADVERTISERS

Bradley T. Shapiro

with Nils Wernerfelt, Anna Tuchman & Robert Moakler



The University of Chicago Booth School of Business



Goals

- Inform beliefs about digital ad effectiveness
 - Massive scale randomized ad experiments.
 - Near universe of advertisers.
 - Incrementality + Generalizability
- How it depends on available data for targeting
- Three thoughts on privacy policy



Digital Advertising on Meta

Budget & schedule

Budget ⓘ

Daily Budget ▼ \$250.00 USD

You'll spend up to \$312.50 on some days, and less on others. You'll spend an average of \$250.00 per day and no more than \$1,750.00 per calendar week. [Learn more](#)

Schedule ⓘ

Start date

Oct 27, 2022

🕒 10:00 PM

Pacific Time

End - Optional

☐ Set an end date

Choose a campaign objective

- ☐ Awareness
- ☐ Traffic
- ☐ Engagement
- ☐ Leads
- ☐ App promotion
- ☒ Sales



Sales

Find people likely to purchase your product or service.

Good for:

- Conversions ⓘ
- Catalog sales ⓘ
- Messenger and WhatsApp ⓘ
- Calls ⓘ

Audience

Define who you want to see your ads. [Learn more](#)

Create new audience

Use saved audience ▼

Custom audiences

Create new ▼

🔍 Search existing audiences

Exclude

Locations

Location:

- United States

Age

20 - 30

Gender

All genders

Detailed targeting

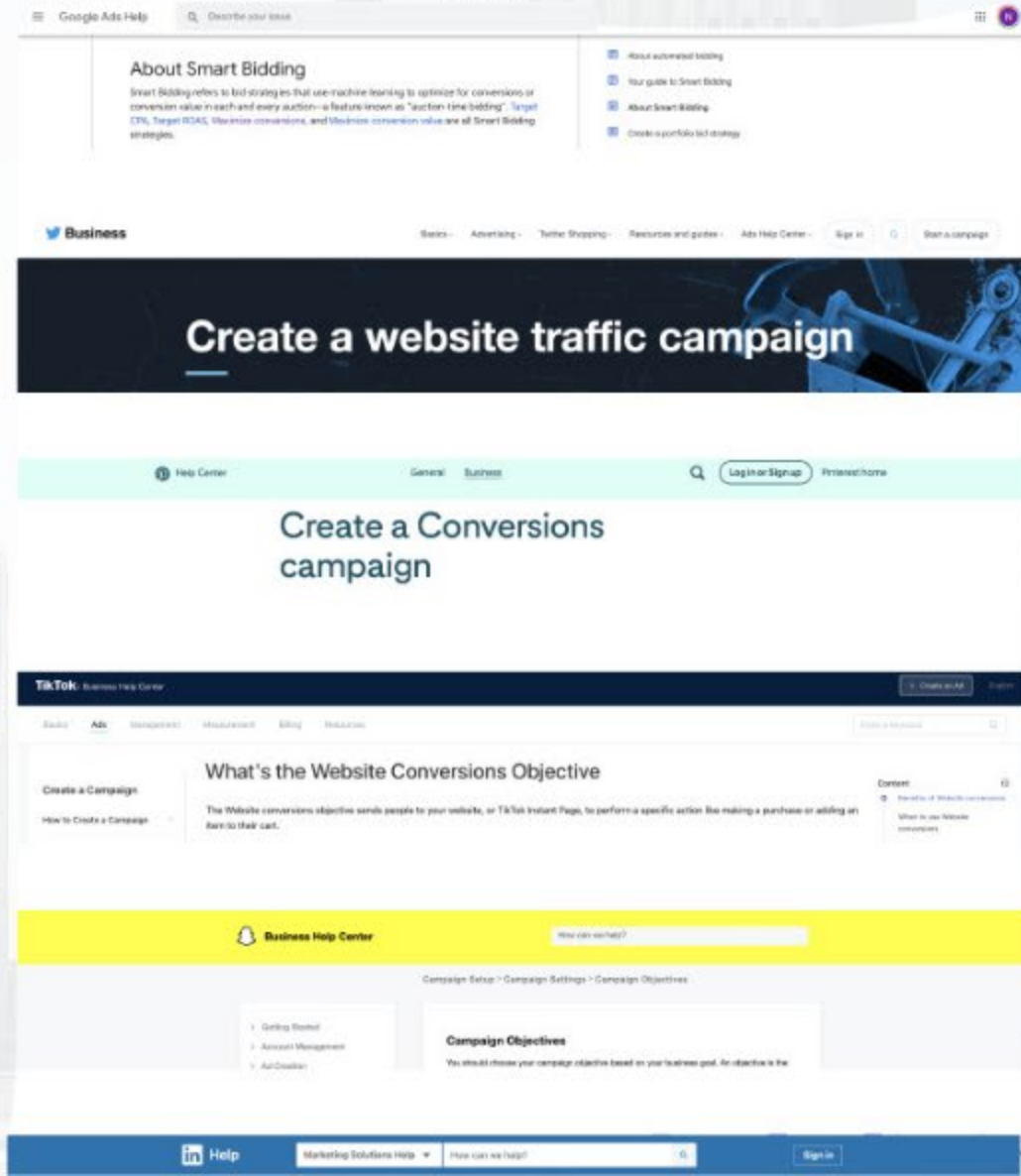
Include people who match ⓘ

Interests > Business and industry

Marketing

Similar on All Digital Platforms

- Google
- Twitter
- Pinterest
- TikTok
- Snapchat
- LinkedIn
- ...



Typical Advertiser



Niche Segment

Unsophisticated

Few employees

No physical store

Ad Budget

< \$100k/year

< \$2k/week

How does “Optimization” Work?

- Outcome observed.
 - Purchase on a website, click on an ad, etc.
- Users generating that outcome enter “training sample”
- AI/ML model identifies users similar to those, serves ad
- AI/ML model monitors and adjusts if subsegments of initial target audience appear unresponsive.

What Data Inform These Algorithms?

- On site (or `first party') data
 - Created on, hence owned by, the platform.
 - E.g. Browsing and **clicks** on facebook
- Offsite (or `third party') data
 - Created away from from the platform
 - E.g. **Purchases** or browsing on advertiser website

data on Google platform (e.g Chrome), mostly first party.

Meta uses meta pixel to match the profile to the person browsing by having the cookie/code.

How Does Meta Get Purchase Data?

- Meta Pixel
 - Piece of code installed by advertiser on website
 - User hits website, pixel fires, cookie data relayed to Meta.

```
<!-- Facebook Pixel Code -->
<script>
  !function(f,b,e,v,n,t,s)
  {if(f.fbq)return;n=f.fbq=function(){n.callMethod?
  n.callMethod.apply(n,arguments):n.queue.push(arguments)};
  if(!f._fbq)f._fbq=n;n.push=n;n.loaded=!0;n.version='2.0';
  n.queue=[];t=b.createElement(e);t.async=!0;
  t.src=v;s=b.getElementsByTagName(e)[0];
  s.parentNode.insertBefore(t,s)}(window, document,'script',
  'https://connect.facebook.net/en_US/fbevents.js ');
  fbq('init', 495591850793067);
  fbq('track', 'PageView');
</script>
<noscript><img height="1" width="1" style="display:none"
```


An Example



Buys



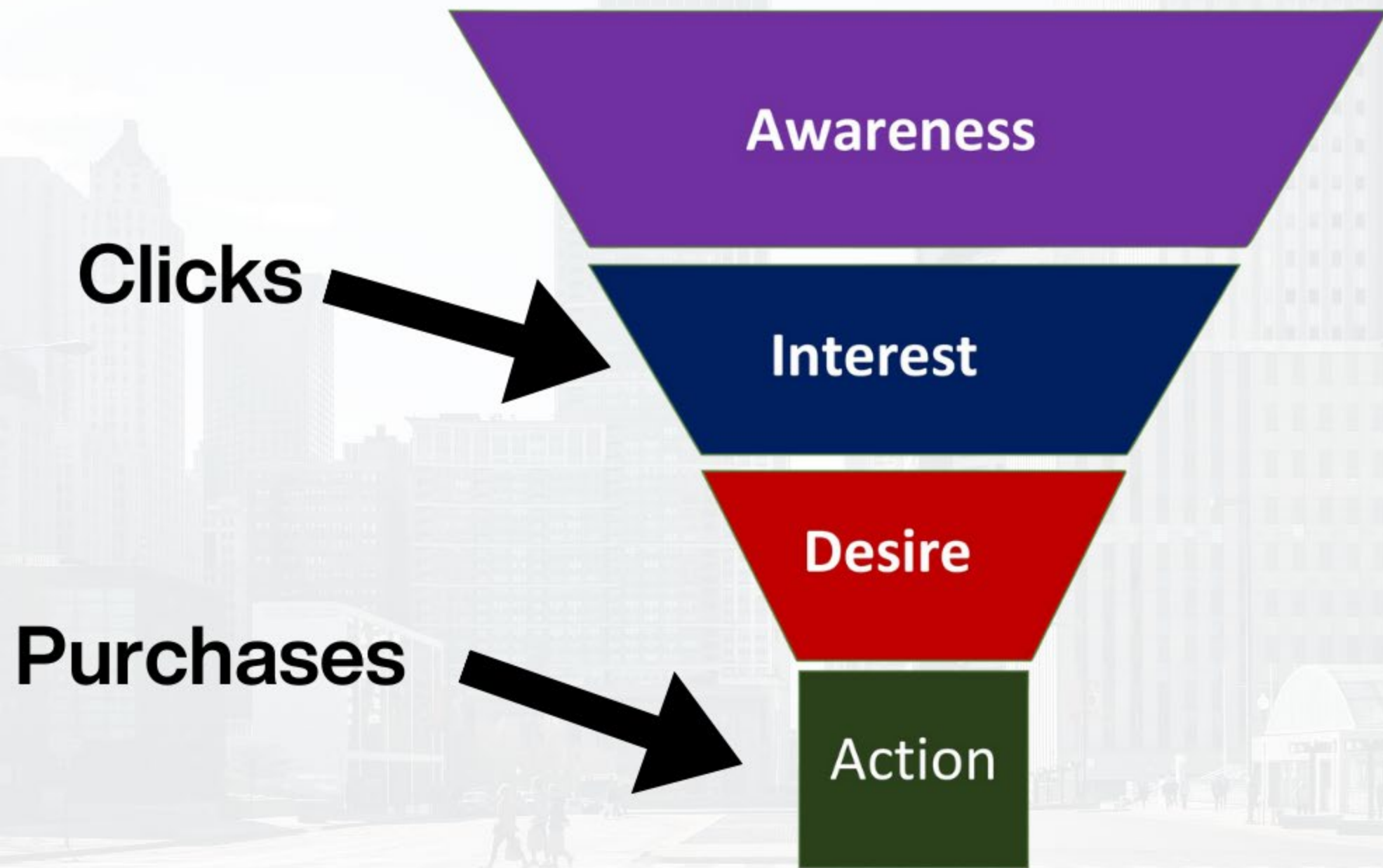
Pixel



Serves ad to



Purchases vs. Clicks



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Experimental Design

5% of target audience,
5% of budget

Purchase Optimized, Exposed

90%

Purchase Optimized, Holdout

10%

5% of target audience,
5% of budget

Click Optimized, Exposed

90%

Click Optimized, Holdout

10%

If Meta don't have access to third party data, they are targeting a potentially wholly wrong group.

Holdout - Second Highest Bidder



Quigley

\$0.55

SRF | 極
SNAKE RIVER FARMS.

\$0.43

wine.com

\$0.41

Treatment JP

Control JP



experiment;
intentionally show
CONTROL JP an ad on
snake river farms to see
how much Control JP
purchase.



\$0.43



SNAKE RIVER FARMS.

\$0.41

It's just not that easy

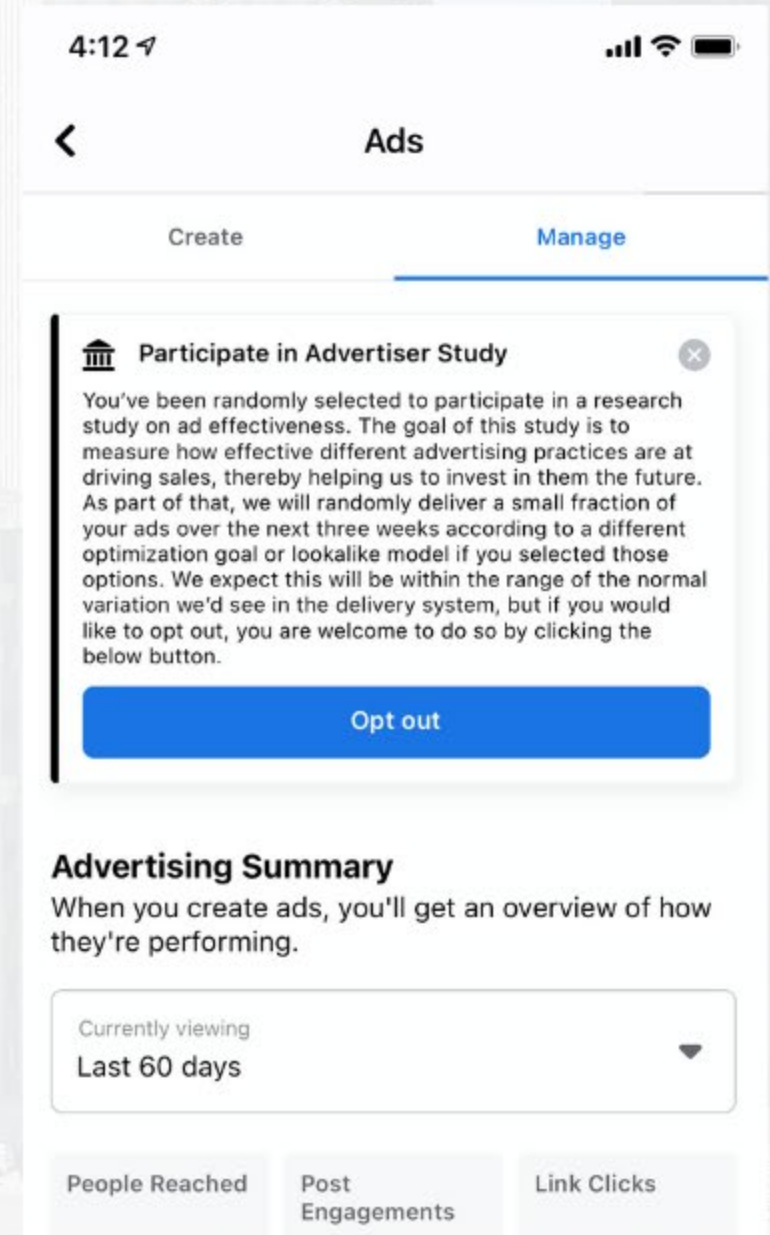
- People like me tell you “just run experiments”
 - Typical advertiser spends $< \$2k$ / week.
 - Each experiment will be under-powered.
- This is where we come in.
 - Learn across many representative advertisers
 - Outcome will be posterior belief distribution

Goals

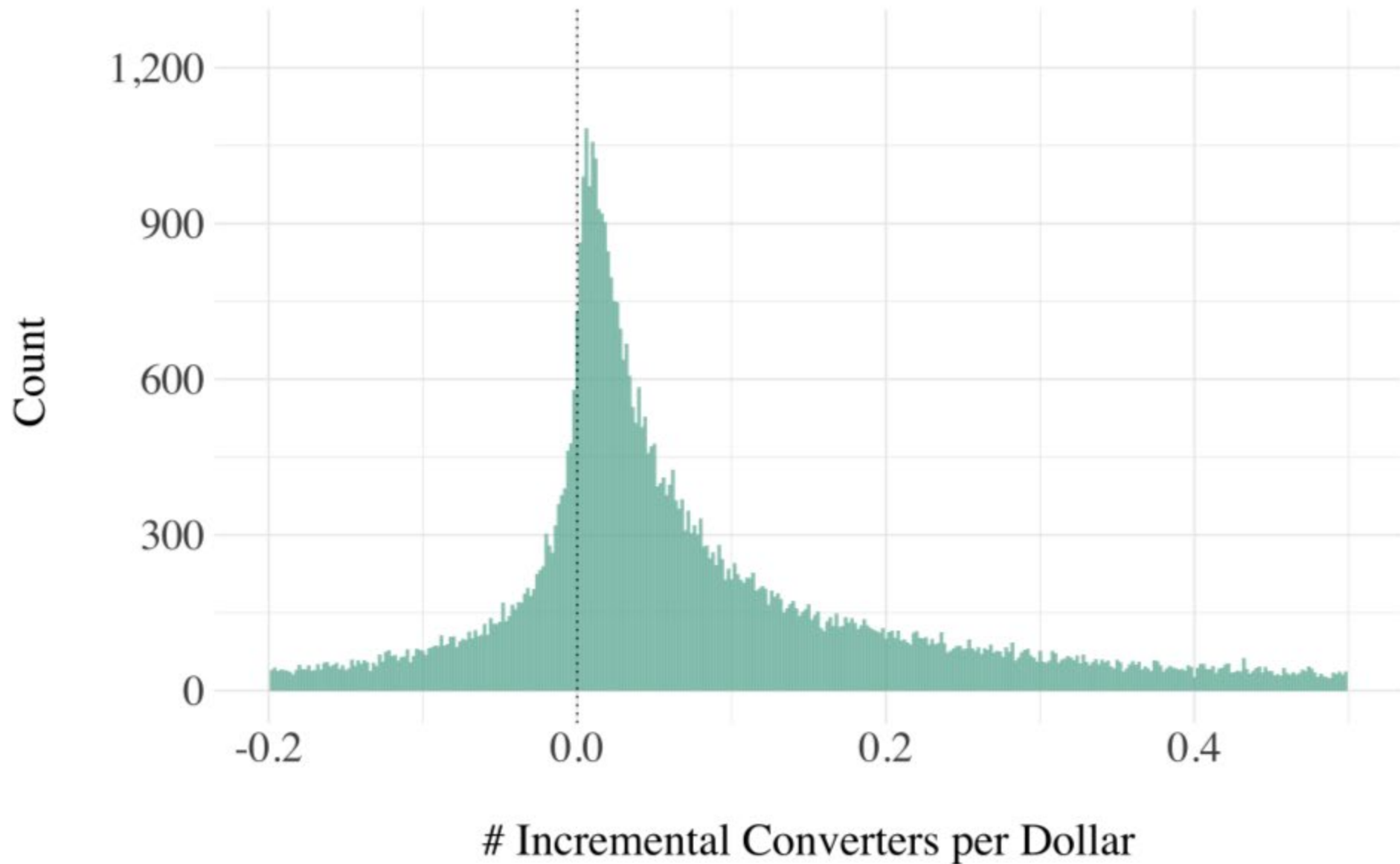
- Inform beliefs about digital ad effectiveness
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 - **Near universe of advertisers.**
 - Incrementality + **Generalizability**
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Sample - Near Universe

- Sent opt-out notice to near universe of advertisers who used offsite conversion optimization in three months prior to experiment (31 languages)
- High percentage did not opt out (94%).
- After cleaning, left with 70,909 experiments
 - Note: large number are using 'incorrectly'



Experimental Treatment Effects



Empirical Bayes Deconvolution

$$\Theta_i \stackrel{\text{iid}}{\sim} g(\theta)$$

Define a grid $\{\theta_1, \theta_2, \dots, \theta_m\}$ of values for the support of g

Let

$$X_i \stackrel{\text{ind}}{\sim} p_i(X_i | \Theta_i)$$

$$g(\alpha) = \exp(Q\alpha - \phi(\alpha)),$$

- Q is an $m \times p$ basis matrix for natural cubic splines with $\text{df} = p$
- $\alpha \in \mathbb{R}^p$ are the weights on the spline basis matrix
- $\phi(\alpha) = \log \sum_{j=1}^m \exp(Q_j^T \alpha)$

Assume $p_i \sim N(\Theta_i, \hat{\sigma}_i^2)$, then letting g_j denote the j th component of g , the marginal probability for X_i becomes

$$f_i(\alpha) = \sum_{j=1}^m p_i(X_i | \Theta_i = \theta_j) g_j(\alpha)$$

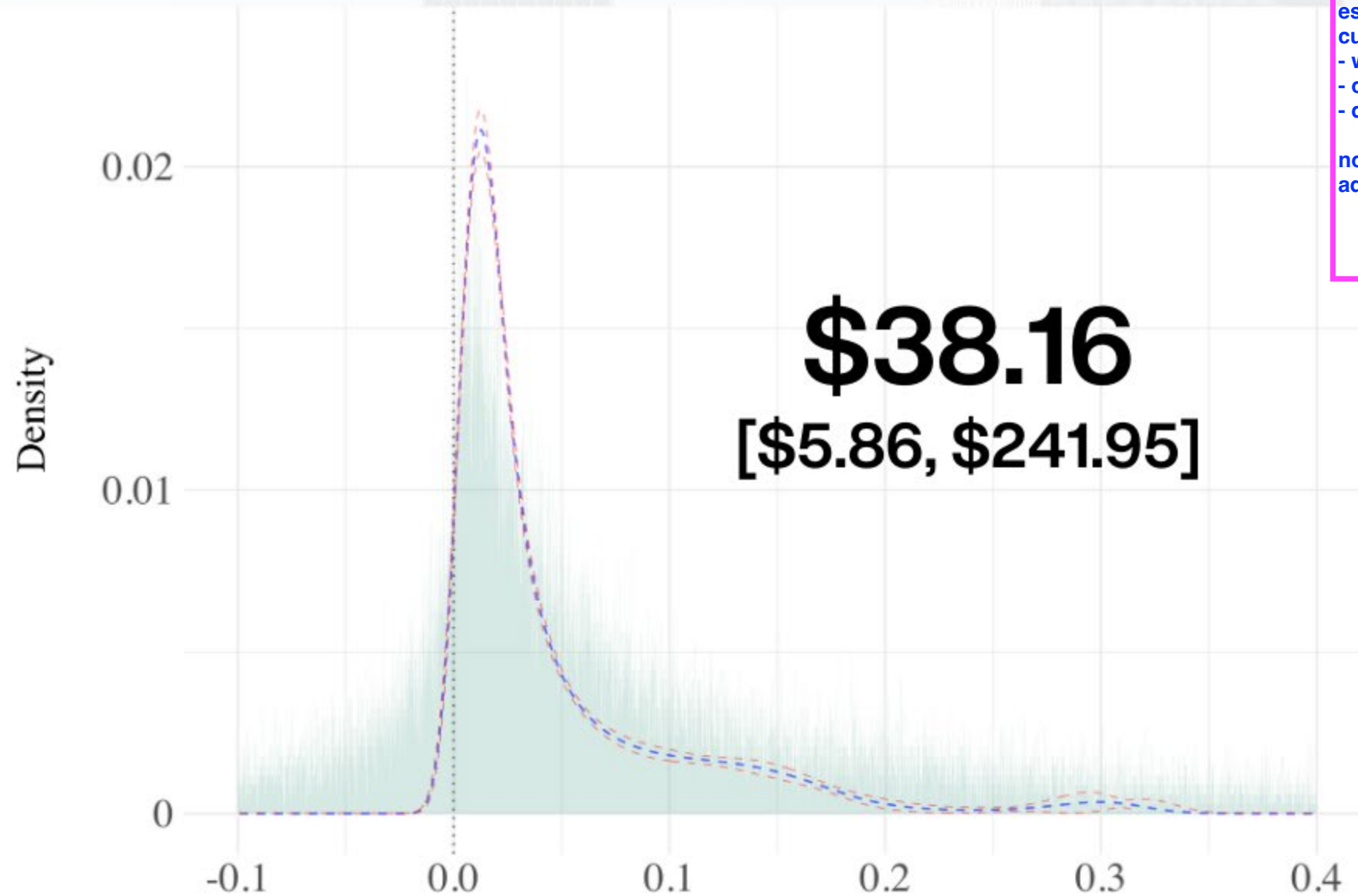
Can then solve via maximum likelihood for a fixed p ; run over a wide range of p 's, and perform model selection via Bayesian Information Criterion

In English

We are doing some (legit statistics) stuff to learn across experiments, decrease the noise from small experiments, and estimate the “true” distribution of advertising effects.

Efron (2016)

Estimated Distribution



estimated \$38 to get a new customer. Is it worth? depends on;
- what is CLV?
- our revenue, margins?
- durable or non-durable product?

note, this is purchase-optimised ads.

The background is a faded, grayscale image of a city street scene. It features several tall skyscrapers in the background, a street with a crosswalk in the foreground, and a subway entrance with a glass canopy on the right side. A few small figures of people are visible on the street, and traffic lights are also present.

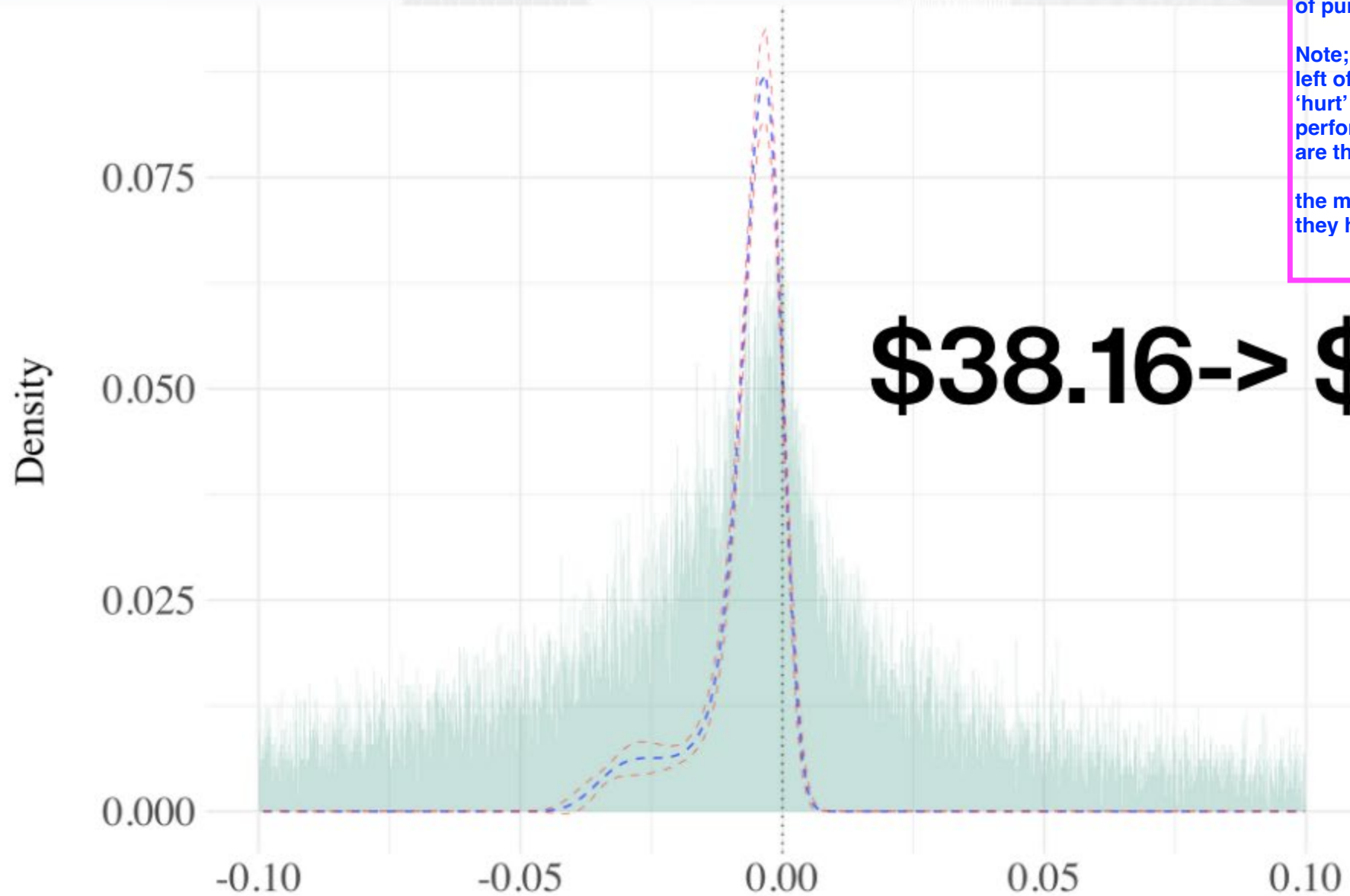
Is it worth it?

It depends...

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- **How it depends on available data for targeting**
 - Importantly, can no longer use purchases. Best we can do is clicks.
- Three thoughts on privacy policy

Purchases vs. Clicks



40% increase in cost (from 38 to 50) if we use click optimisation instead of purchase optimisation.

Note; what is the secondary hump left of peak? some advertisers are 'hurt' really bad due to poor performance (poor targeting). who are they?

the more niched an advertiser is, they higher chance of being 'hurt'.

\$38.16 -> \$49.93

Larger Scale



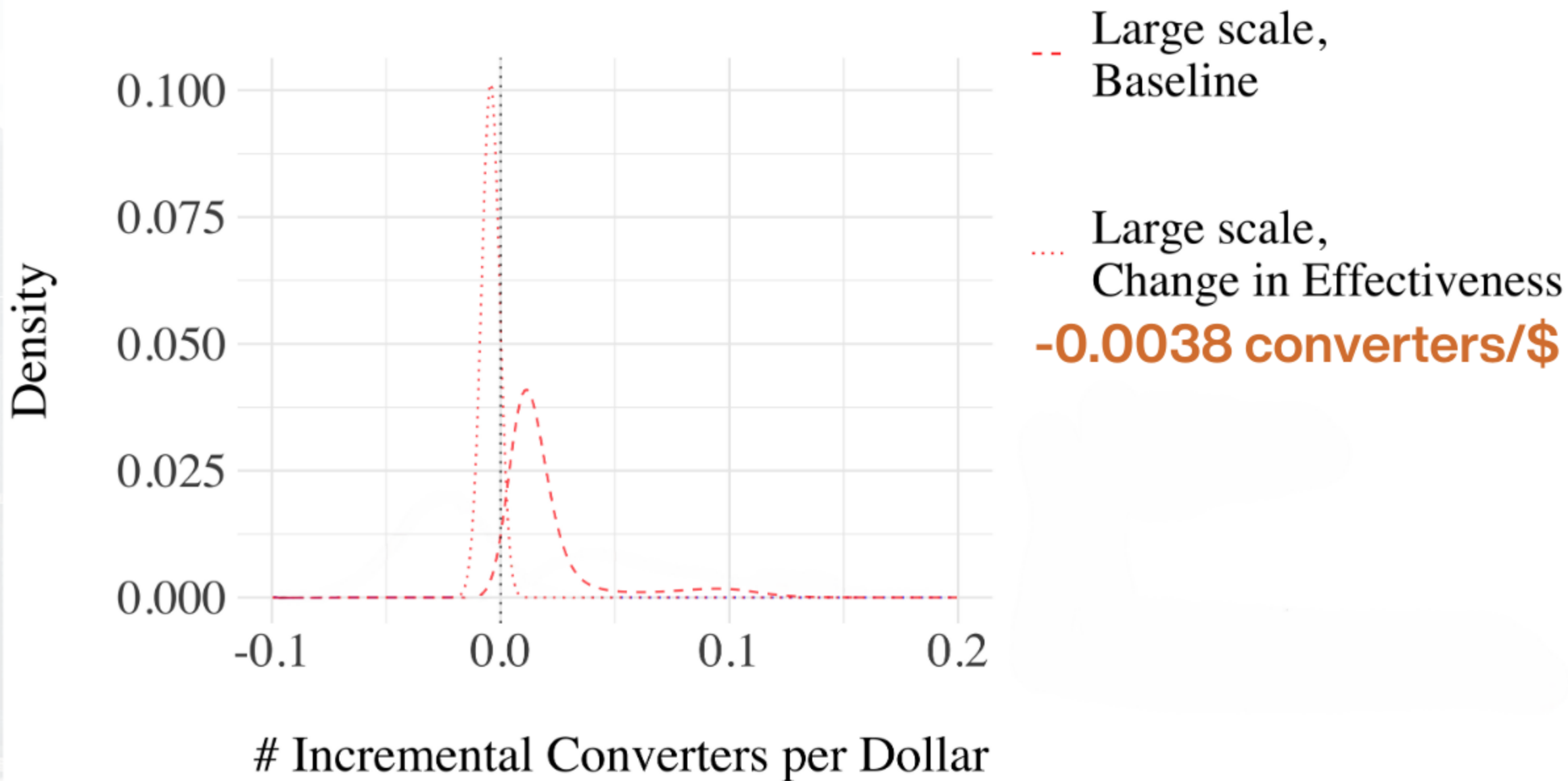
VIVOBAREFOOT

vs

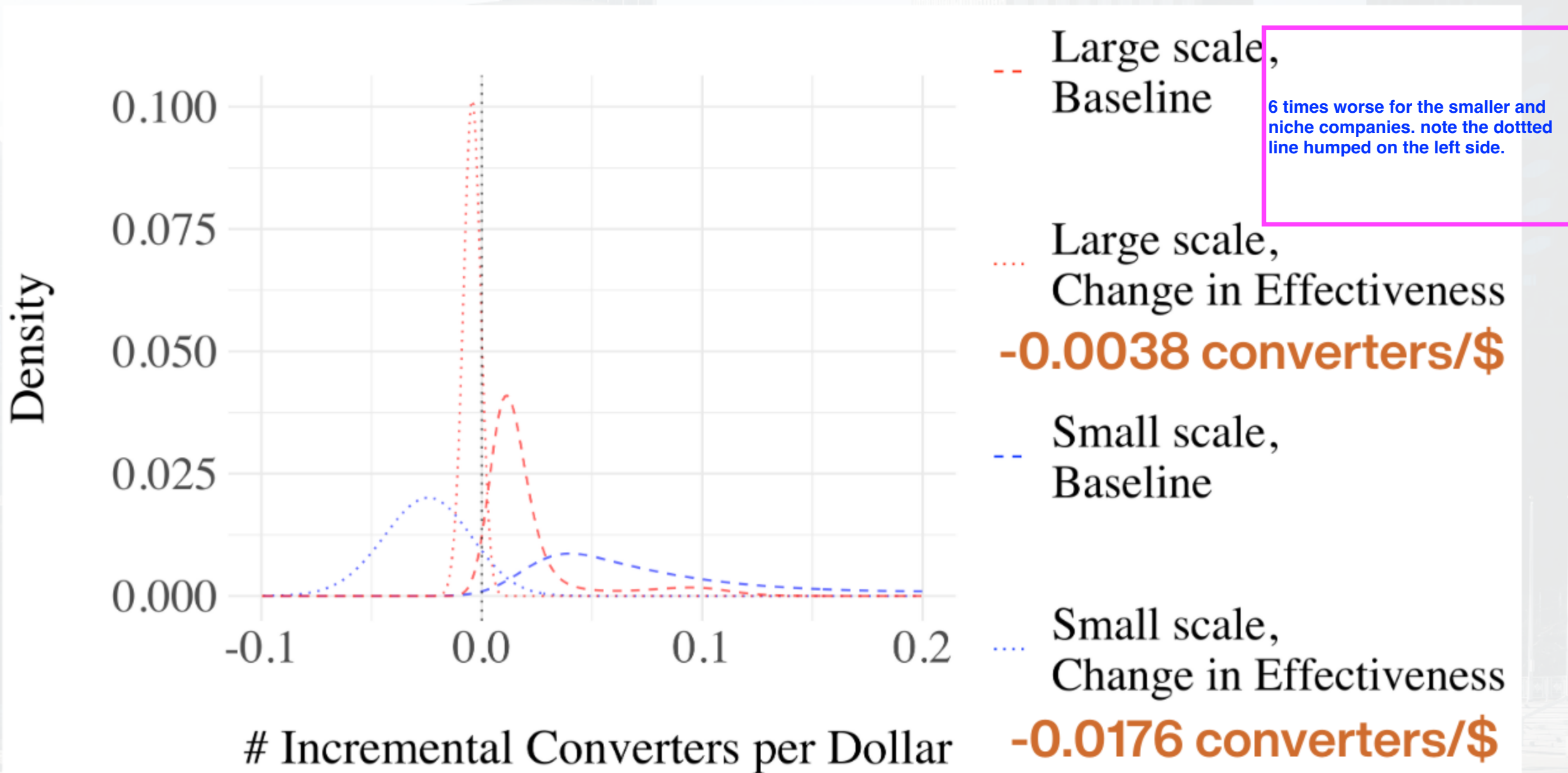


Smaller Scale

Large vs. Small Scale Advertisers



Large vs. Small Scale Advertisers



What Should You Do?

if you are smaller niched advertiser,

- Consider your choice of ad platform.
- Consider adjusting your bids.
- Differences big vs. small.

Is Using Offsite Data Good for Customers?

Good

- Find products they like
- Repeat purchase
- Helps them meet needs

Bad

- Unobserved low quality
- One time purchase
- Hooked onto Cigarettes?

Purchase Optimization

**\$112.69/ 'long term'
customer**

Click Optimizaiton

**\$154.77/ 'long term'
customer**

Better data



Better Matches

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separate note; privacy policy.

What is the rationale of GDPR to have data privacy? some people will be manipulated to pay more when their data is revealed. (usually these people have higher WTP and/or better off).

studies show, most people who care about their own privacy are correlated to their wealth level.

Privacy Policy I - Platform Competition



Ad prices going up? Quality going down?

Privacy Policy II - Product Competition



vs



Lower innovation? Less variety? Higher prices?

Privacy Policy III - Property Rights?



Is there room for a compromise?

Wrapping Up

- We ran over 70,000 ad targeting experiments.
- Purchase optimization - \$38/ incremental customer
 - Much less for smaller scale companies.
- Click optimization - +40%/incremental customer
 - Worse for smaller scale companies
- Lots of variance in both. Our results a useful prior.

Questions?